

555.644 Introduction to Financial Derivatives
Final Exam

Consider a European call option on a non-dividend-paying stock; when the option is written, the stock price is S_0 , the volatility of the stock price is σ , the strike price is K , the continuously compounded risk-free rate is r , and the term to expiration is T ; let c be the price of the option. The Black-Scholes formula for the option price is

- A. $c = S_0 N(d_1) + Ke^{-rT} N(d_2)$
- B. $c = S_0 N(d_1) - Ke^{rT} N(d_2)$
- C. $c = S_0 N(d_1) - Ke^{-rT} N(d_2)$
- D. $c = S_0 N(d_2) - Ke^{-rT} N(d_1)$

where $N(x)$ is the cumulative probability distribution function for a standardized normal distribution and d_1 and d_2 are parameters dependant on the structure of the option, the level of interest rates, and the volatility of the stock price.

13. (a) Using the terminology of the last question (re-printed above – Question 8 from the multiple choice section), specify the Black-Scholes formula for the price of a European put option on a non-dividend-paying stock

(b) Explicitly describe the relationship of the parameters d_1 and d_2 to the structure of the option, the level of interest rates and the volatility of the stock price and the relationship of the parameters to each other; use the notation of the last question (e.g., write the formulas for d_1 and d_2)

(c) We found that for a dividend yielding stock that there was a simple enhancement possible to convert the result in question 3) to the case of a European call option on a dividend yielding stock. Let q represent the dividend yield and describe the enhancement (which we found appropriate in many representations). Also, show the result for the European call option on a dividend yielding stock (include the formula for d_1 and d_2).

14. For the stock price process $dS = \mu S dt + \sigma S dz$ where dz is the Weiner process, μ is the expected rate of return on the stock and σ is the volatility of the stock price; suppose f is the price of a derivative that is contingent on S ; the Black-Scholes-Merton differential equation governs all the derivatives that can be defined with S .

- a) Show the Black-Scholes-Merton differential equation for f
- b) What is the boundary condition of this equation for a European put option when $t = T$.

15. For the Black-Scholes-Merton differential equation we see the property that it does not involve any variables that are affected by the risk preferences of investors. The consequence of this is that the equation has solutions that adhere to risk-neutral principles.

- a) What is the Principle of Risk Neutral Valuation and what is the procedure that leads to a risk-neutral valuation?
- b) Apply the risk-neutral procedure to value a long forward contract on a non-dividend paying stock with price S , that matures at time T , and with delivery price K .
- c) Show that this result satisfies the Black-Scholes-Merton differential equation.

16. A futures price is currently 50. At the end of 6 months it will be either 55 or 45. The risk free interest rate is 6% per annum. We want to value a 6-month European call option with a strike price of 50.

a) What is the Black-Scholes equation to govern this situation?

b) What is the value of the option? (Consider using a one-step binomial tree)

17. Consider a stock index currently standing at 250. The dividend yield, q , on the index is 4% per annum and the risk-free rate, r , is 6% per annum. A 3-month European Call option on the index with a strike price of 245 is currently worth \$10. We want to value a 3-month European put option on the index with a strike price of 245.

a) The put-call parity relationship for European-style options on a non-dividend paying stock with price S_0 and strike K is $c + Ke^{-rT} = p + S_0$, where r is the risk-free rate and c & p are the value of the call and the put, respectively. What is the put-call parity relationship for the *dividend* yielding index? Carefully define your notation.

b) What is the value of the put in this question?

18. A futures price is currently 40. It is known that over *each* of the next *two* 3-month periods it will either rise by 10% or fall by 10%. The risk free interest rate is 8% per annum. (Consider using a 2-stage binomial tree for these questions.)

a) What is the value of a 6-month European call option on the futures with a strike price of 40?

b) If the call were American, would it ever be worth exercising early?

c) What is the value of a 6-month European put option on the futures with a strike of 40?

d) If the put were American, would it ever be worth exercising early? What would its value be?